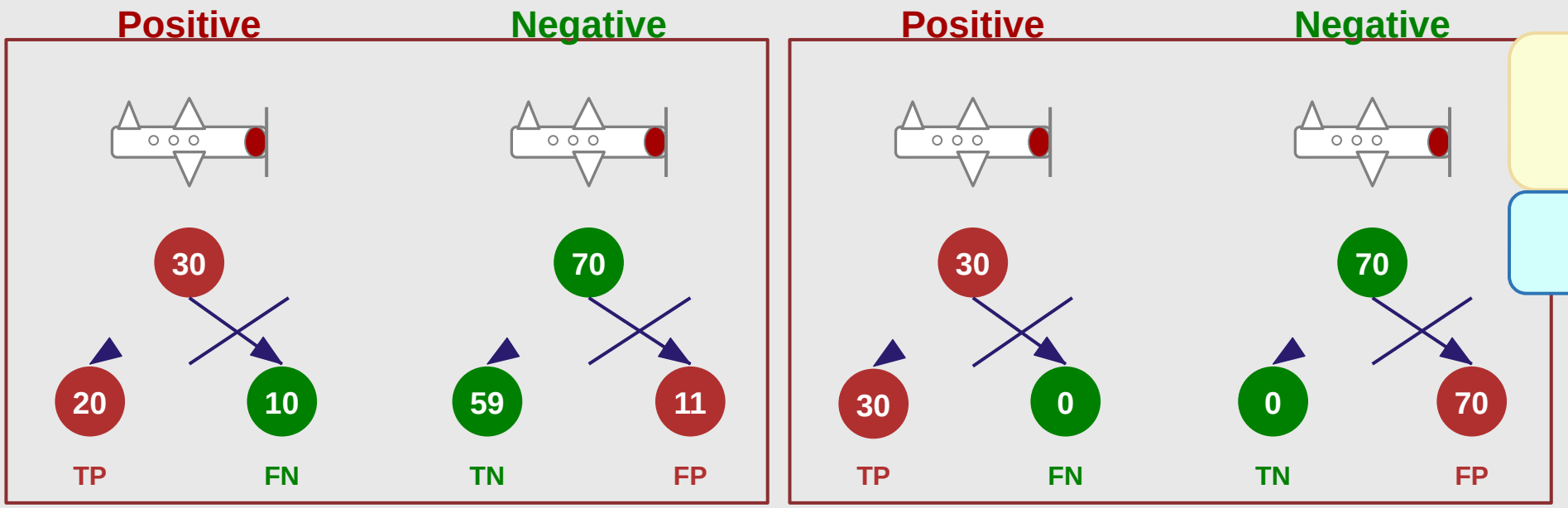
Combine Precision and Recall in one number



Precision

$$\frac{TP}{TP + FP}$$

Recall

$$\frac{TP}{TP + FN}$$

$$\frac{1}{\text{F1 Score}} = \frac{1}{2} \times \left(\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}\right)$$

Penalize Small Value (X) by Using 1/X
Normalize (by * 0.5) to get F1 Score between 0,1

Precision= 20/(20+11) =64.5%

Recall= 20/(20+10) =66.6%

F1 Score= 68.7%+60% =**65.55%**

Precision= 30/(30+70) =30.3%

Recall= 30/(30+0) =100%

F1 Score= 30.3%+100% =**46%**